EX.NO:1
DATE: 17.7.2026

Basic Networking commands in Linux and windows operating system

AIM:

To study and execute various Linux commands for file and directory management text processing and basic networking operations

→ study of various commands used in Linux and

description: prints the current working directory

output:

$ pwd

/home/linuxuser

mkdir

description: creates a new directory

output

$ mkdir fedora

$ ls

Desktop Documents Downloads fedora cd

description: change the current directory

output: $ cd fedora

$ pwd

/home/linuxuser/fedora

touch title name.txt

description creates an empty file

output:

$ touch one example.txt

$ ls

example.txt

echo "Hi" > filename.txt

description: writes text into a file

output:

$ echo "Hi" > example.txt

$ cat file name.txt

Hi

grep

description: Searches for the specified pattern in a file.

output $ grep "Hi" example.txt

. the me

description: display the no. of lines, words & bytes in a file

output:

$ wc example.txt

9 13 example.txt

wc -l

description: counts the no. of lines in a file

output:

$ wc -l example.txt

1 example.txt

wc -w

description: counts the no. of words in a file

output

$ wc -l example.txt

3 example.txt

wc -l

description : count the no of characters in a file

output :

$ wc -m example.txt

3 example.txt

wc -L

description : display the length of the longest line in a file

output :

$ wc -l example.txt

2 example.txt

Basic networking commands

host name

description : display the routing table

output

$ route

Kernel IP routing table

netstate

description : Show network connection and routing information

output :

$ net stat

Active Internet connection

ip address show

description : test connectivity to a host &
its network interface details

output

.9 ip address show

1: lo

2: enp0s3

no lookup

description: finds the ip address of domain

output: -

$ nslookup www.google.com

name: www.google.com

address: 142.250.xxx.xx x

ifconfig

description: display network and enforce configuration

output:

inet config

enp0s3

inet 10.0.2.15

ping

description: test connectivity to a host.

output:

ping www.google.com

64 bytes from...

arp

description: display the ARP cache

output:

arp -a

- gateway (10.0.2.2)

traceroute

description: display the route taken by packets to reach a destination

output:

$ traceroute www.google.com

1- gateway

2...

Result: 17/7/21

Thus the linux & windows commands are studied & executed successfully

Ex.No: 2
Date
23.7.26

Learning and Assignment of IP address manually to computers

Aim:

To learn IP addressing and manually assign an IP address to a computer. To verify the assigned IP address using Linux networking commands

Theory:

An IP address is a unique numerical address assigned to every device connected to a network It enables communication between devices over a network An IPv4 address consist of four octets separated by dots.

An IP address can be assigned can be assigned either manually or automatic using DHCP.

If manually assigned IP address remain unless changed by the user Along with the IP address parameters such as the subnet mask, default gateway and DNS server also configured for proper network communication

In Linux the ip command is used to display and manage IP address, network interfaces routing tables and neighbouring devices It is one of the most commonly used commands for network configuration and troubleshooting

commands and sample output

1) display all IP address

commands:

- ip address show

example output:

2: ens1a:

inet 172.16.211.128/24

inet6 fe80:20c:29ff:fe75:9927/64

observation:

display the IPV4 and IP V6 address of all networks

2. display a specific network interface.

command:

ip ~~addr~~ show dev ens160

sample output:

2: ens160:

inet 172.16.211.128/24

3: Display Routing Table

command

ip route show

Sample output:

default via 172.16.211.2 dev ens160

172.16.211.0/24 dev ens160

observation:

Displays the routing table and the gateway

4. Display the neighbour table:

command

ip neigh show

Sample output

172.16.211.1 REACHABLE

172.16.211.234 STABLE

Observation:

Display the neighbouring devices and their MAC address

RESULT:

The IP address was manually assigned and verified successfully using linux networking commands

EXP: 3
DATE: 23.7.26

Study of different types of Network cables and crimping of cable with RJ45 connector

Aim:

To study different types of network cables and understand the crimping process using an RJ45 connector

Concept:

Network cables are used to connect computers and other networking devices for data communication the most commonly used cables in LANs is the unshielded twisted pair cable. An RJ45 connector is attached to the ends of the cable to establish a network connection are arranged according to the T568A or T568B wiring standard before crimping and the completed cable is tested to ensure proper connectivity

Types of Network cables

	Cable Type	Description
1	UTP (unshielded Twisted Pair)	Commonly used for Ethernet LAN's and office network
2.	STP (shielded Twisted Pair)	Used in area with high electrical and electromagnetic interface
3.	Coaxial cable	Used for cable television, broadbanded internet and CCTV systems
4.	Optical Fiber Cable	Used for high speed high Bandwidth and long distance communication

Crimping procedure:

1. Cut the UTP cable to the required length.
2. Strip about 2-3 cm of the outer insulation
3. Untwist and straighten the wires
4. Arrange the wires according to the T568B standard
5. Twist the wires ends evenly
6. Insert the wires into the RJ45 connector
7. Crimp the connector using a crimping tool
8. Test the cable using a LAN cable tester

Observation:

* The wires were arranged according to the T568B standard
* The RJ45 connector was crimped successfully.
* The cable Tester indicated proper continuity
* The cable was ready for network communication

Result:

The different types of network cables were studied and the UTP cable was successfully terminated using and RJ45 connector and tested for proper connectivity

EX NO: 4

DATE:

IMPLEMENT PACKET SNIFFING IN WIRE SHARK

Aim:

To capture and analyze various network protocol packets using wireshark and apply display filters to inspect them

Software used:

* Wireshark Network Protocol Analyzer

Procedure:

1. Open wireshark
2. Select Local Area Connection.
3. Go to Capture → Options.
4. Set: Stop capture automatically after 100 packets
5. Click Start to begin capturing
6. Using filter bar to type specific protocols like tcp, arp etc..
7. Use Statistics → Flow Graph to view packet flow
8. Click File → Save to store the capture packets

Task & output

1. Filter: Display TCP/UDP Packets and Flow graph

   ★ Filter used TCP or UDP

   ★ output: displayed only TCP/UDP Packets

   ★ Flow graph is viewable from statistics -> Flow Graph

2. Filter: display only ARP Packets

   ★ Filter used: arp

   ★ output: displayed only Address resolution Protocol Packets

3. Filter display only DNS Packets and Flow graph

   ★ Filter used dns

   ★ output displayed only DNS query and response

   ★ Flow graph: show DNS request response communication

4. Filter: display only HTTP packets

   ★ Filter used: http

   ★ output: captured HTTP traffic

5. Filter: display ICMP Packets

   ★ Filter used icmp or ip

   ★ output: shows ICMP packets and IP packet info

6. Filter: display only DHCP Packets

   ★ Filter used: bootp

   ★ output: display DHCP discover, offer, Request, Acknowledgement packets

RESULT:

Successfully captured and filtered different protocol packet using wireshark and understood how to inspect, and analyse them with tools like filters and flow graphs.

Exp No:5
Date 24.7.26

Develop a customize ping command
To test server connectivity

Aim:

To develop and execute customized ping commands to test the connectivity, response time and packet transmission between a client

Theory:

. The ping command is a network utility used to verify whether a remote host or server is reachable over an IP network it uses the Internet control message protocol echo request and echo reply message to check the connectivity

. A customized ping command allows the user to specify parameters such as.

★ Number of echo requests
★ Packet size
★ Time to live
★ Timout

. A, ping helps in.

★ Testing network connectivity
★ Measuring round trip time
★ Detecting packet loss
★ Troubleshooting network problem

Requirement

- A computer / Laptop with windows OS
- Command Prompt
- Active internet or local area network
- Reachable server

Commands used

Ping google.com

Ping google.com -n 5

Ping google.com -l 1000

Ping google.com -w 2000

Ping 8.8.8.8

Procedure:

1. Open a Command Prompt
2. Type the required Ping command
3. Press enter
4. Observe the reply from the server
5. Note the response time TTL value and Packet loss
6. Repeat using different Ping parameters

Observation: observation:

Command	Observation
ping google.com	Server reachable replies received successfully
ping	
ping google.com -n 5	Five packet sent and received successfully
ping google.com -l 1000	Server respond to 1000 byte packet
ping google.com -w 2000	Respond recieve within specified timeout
ping 8.8.8.8	Successfully communication with google's DNS server

RESULT:

The various ping commands were executed successfully The server connectivity was verified and network performance was observed

EX NO:9

DATE:

DESIGN A SIMPLE TOPOLOGY AND CONFIGURE WITH ONE ROUTER, TWO SWITCHES AND PCS USING CISCO PACKET TRACER

AIM

To design and configure a simple network topology consisting of one router, two switches, and multiple PCs in Cisco Packet Tracer, and enable communication between PCs

Definitions:

★ Router: A networking device that forwards data packets between computer networks, directs traffic on the internet

★ Switch: A device that connects devices within a network and uses MAC address to forward data to the correct destination

★ PC: End devices in a network that user uses interact with send and receive data

★ Cables:

★ Straight through cable: used to connect similar devices like switch to switch

★ Console cable: used to connect PC to router or switches console port for configuration

EQUIPMENT NEEDED:

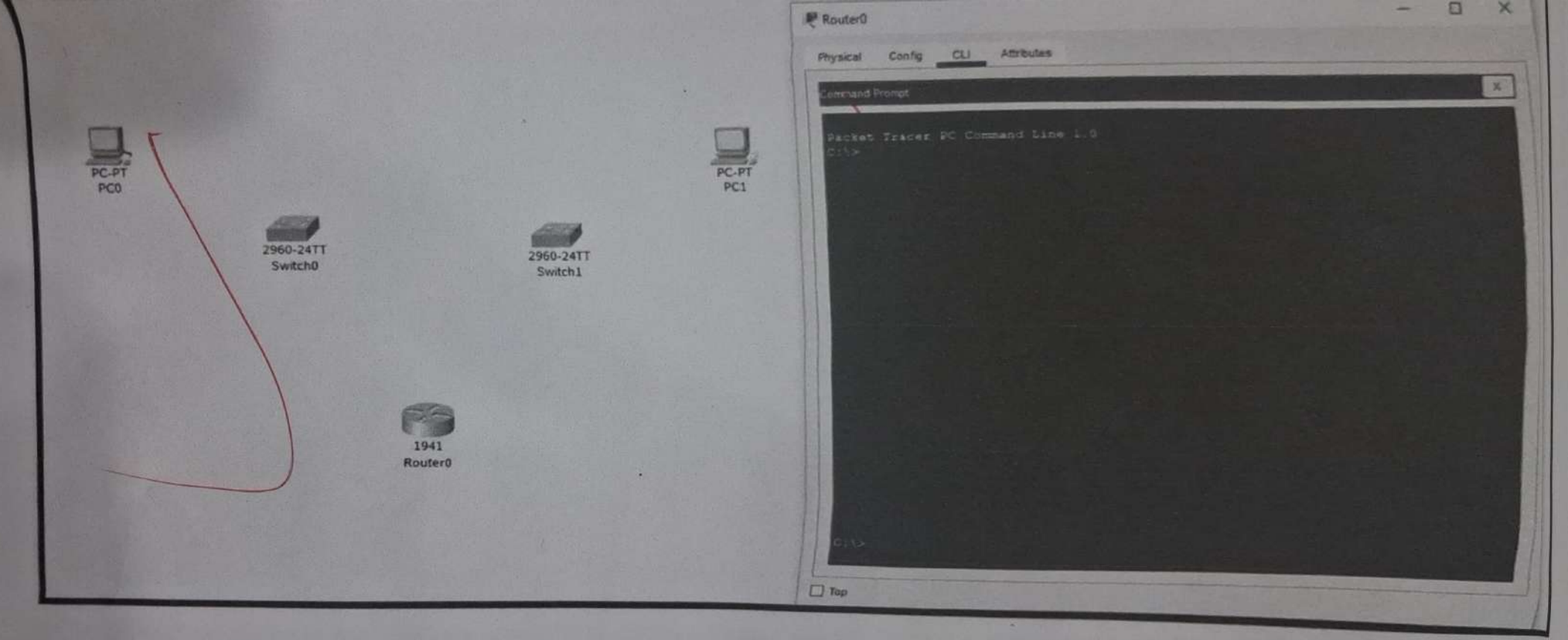
★ CISCO Packet Tracer software

★ 1 Router

★ 2 switches

★ 4 PCs

★ Appropriate cables


PC-PT
PC0
2960-24TT
Switch0
1941
Router0
2960-24TT
Switch1
PC-PT
PC1
Router0
Physical
Config
CLI
Attributes
Command Prompt
Packet Tracer PC Command Line 1.0
C:\>
C:\>
Top

STEP-BY-STEP PROCEDURE:

Step 1: Open Cisco Packet Tracer

★ Launch Cisco Packet Tracer on your computer

Step 2: Add devices to work space

★ From the bottom device toolbar, drag and drop

★ 1 Router (1941)

★ 2 Switches (2960)

★ 4 PCs

Step 3: Connect devices with cables

★ Use copper straight through cable to connect:

★ Router gigabit Ethernet 0/0 to Switch 0 Fast ethernet 0/1

★ Router gigabit Ethernet 0/1 to Switch 1 Fast ethernet 0/1

★ PC 3 to the Switches

★ PC 0 to Switch 0 Fast ethernet 0/2

★ PC 1 to Switch 0 Fast ethernet 0/3

★ PC 2 to Switch Fast ethernet 0/2

★ PC 3 to Switch Fast ethernet 0/3

Step 4: ★ Configure Router interfaces

★ Click the router > CLI tab

★ Enter the following commands

★ terminal interface fa 0/0 192.168.1.1 255.255.255.0 no shutdown exit

shutdown exit and write memory

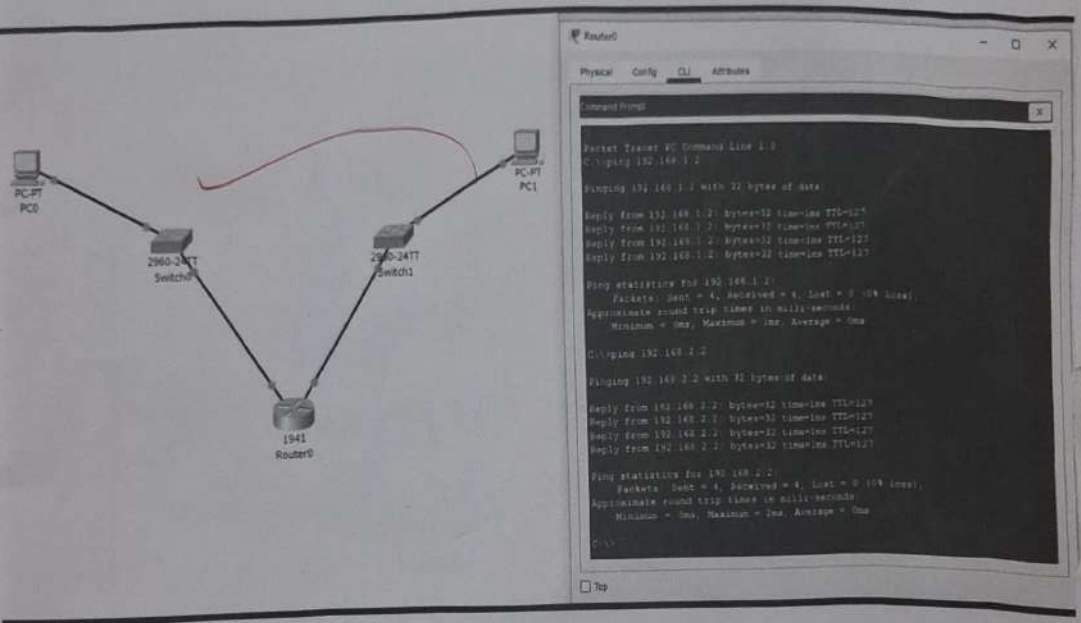
Steps: configure PC IP Address

* click on each PC > desktop tab > ip configuration and assign the following IP's and gateway device IP.

. PC0 192.168.1.10 255.255.255.0 192.168.1.1

Step 6:

* Save and test connectivity.
* Save the packet Tracer File



RESULT:

The PC connected to different switches can communicate with each other via the router. The successful ping replies verify that the network is properly configured and operational

EX NO:10
DATE:

CUSTOMIZE SWITCH WITH NETWORK MODULES USING CISCO PACKET TRACER

Aim:

To learn how to add a network module to a Cisco switch, connect multiple PCs and a server, assign IP address and test connectivity in Cisco Packet Tracer.

Description:

This experiment is about building a simple computer network using a switch with extra module ports. You will connect three PCs and one server, assign IPs and check if all devices can communicate. This helps you understand how modular switches are used to create flexible networks.

DEFINITION:

A network module is an extra component you can add to a switch to give you more ports or add new types of connection. In Packet Tracer, modules such as HWIC-4ESW let you expand the switch's capabilities.

Procedure:

1. Open Cisco Packet Tracer and set up your workstation

2. Drag and drop a 2960-24TT switch to the centre.

3. Add five PCs and the Server-PT from the end devices panel

4. Power off the switch:

- Click the switch open the physical tab and turn of the power button

5. Install "HWIC-4ESW" module on the switch.

- In the physical tab, drag HWIC-4ESW into an empty module slot:

6. Power on the switch and let it boot

7. Connect all devices using copper straight through cables

PC0 -> switch Fast ethernet 0/1

PC1 -> switch Fast ethernet

PC2 -> Switch Module Port 1

PC3 -> Switch Module Port

PC4 -> Switch Module Port3

8. Test connectivity.

INPUT:

Devices:

- 1 Cisco 2960-24TT Switch
- 1 HWIC-4ESW network module
- 5 PC's
- 1 Server PT

Cables:

- 1 copper straight through cable IP configuration
- Use static IP address as listed above

OUTPUT:

A network where every PC and the server can successfully ping each other proving that both cables in and module ports of the switch are functioning

RESULT:

The experiment demonstrate successful customization of a switch with a network module and connectivity between all devices. The network works as expected and all devices communicate

Ex No:7

Date

Develop a simple calculator using XMLRPC

Aim:

To create a Python based FTP scanner using the built in XML-RPC protocol in Python that performs basic arithmetic operations like addition, subtraction, multiplication and division through client-server communication

Description:

XML-RPC is a protocol that uses XML to encode requests and responses and HTTP as a transport mechanism it enables communication between a client and a server where the client and a server where the client and a server where the client invokes methods hosted on the server remotely

In this experiment a simple calculator service is created using XML-RPC in Python

* The server host arithmetic function
* The client connects to the server to perform operations like addition, subtraction, multiplication and division

CODE: Procedure:

* open a Python IDE or terminal

* Create two separate Python files server.py and client.py

* write the provided server side code in server.py and client side code in client.py

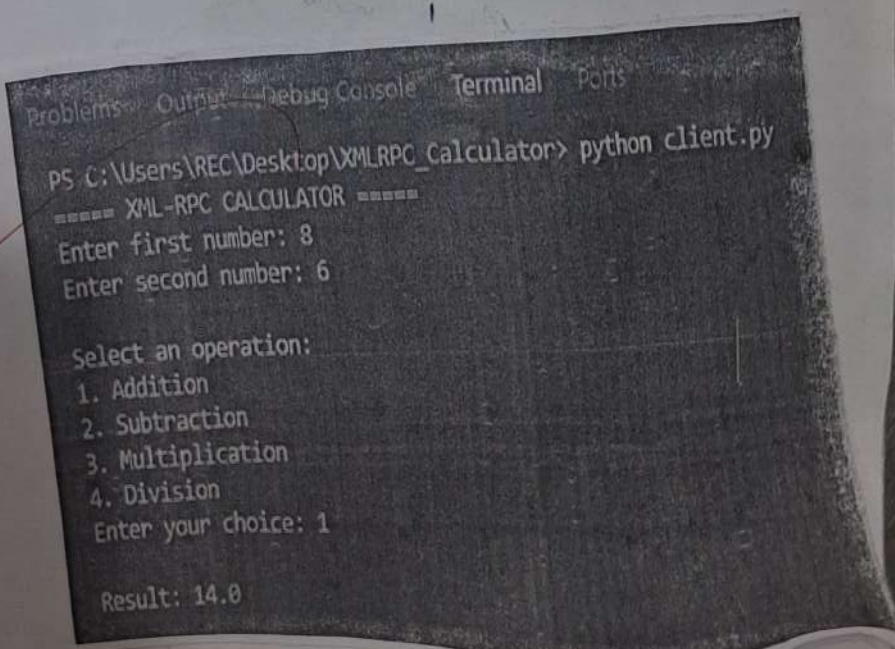
* Run the server.py first to start the XML-RPC server

* Run the client.py file to perform arithmetic operation remotely

* observe the results displayed on the client screen and verify the server output

Software Requirment:

* operating system: windows / linux
* Python version 3.x or above
* Modules Required: xmlrpc.server, xmlrpc.client
* Text Editor or IDE: IDLE / vs code / pycharm



CODE:

Server side code:

```
from xmlrpc.server import SimpleXMLRPCServer

class Calculator:
    def add(self, x, y):
        return x + y

    def subtract(self, x, y):
        return x - y

    def multiply(self, x, y):
        return x * y

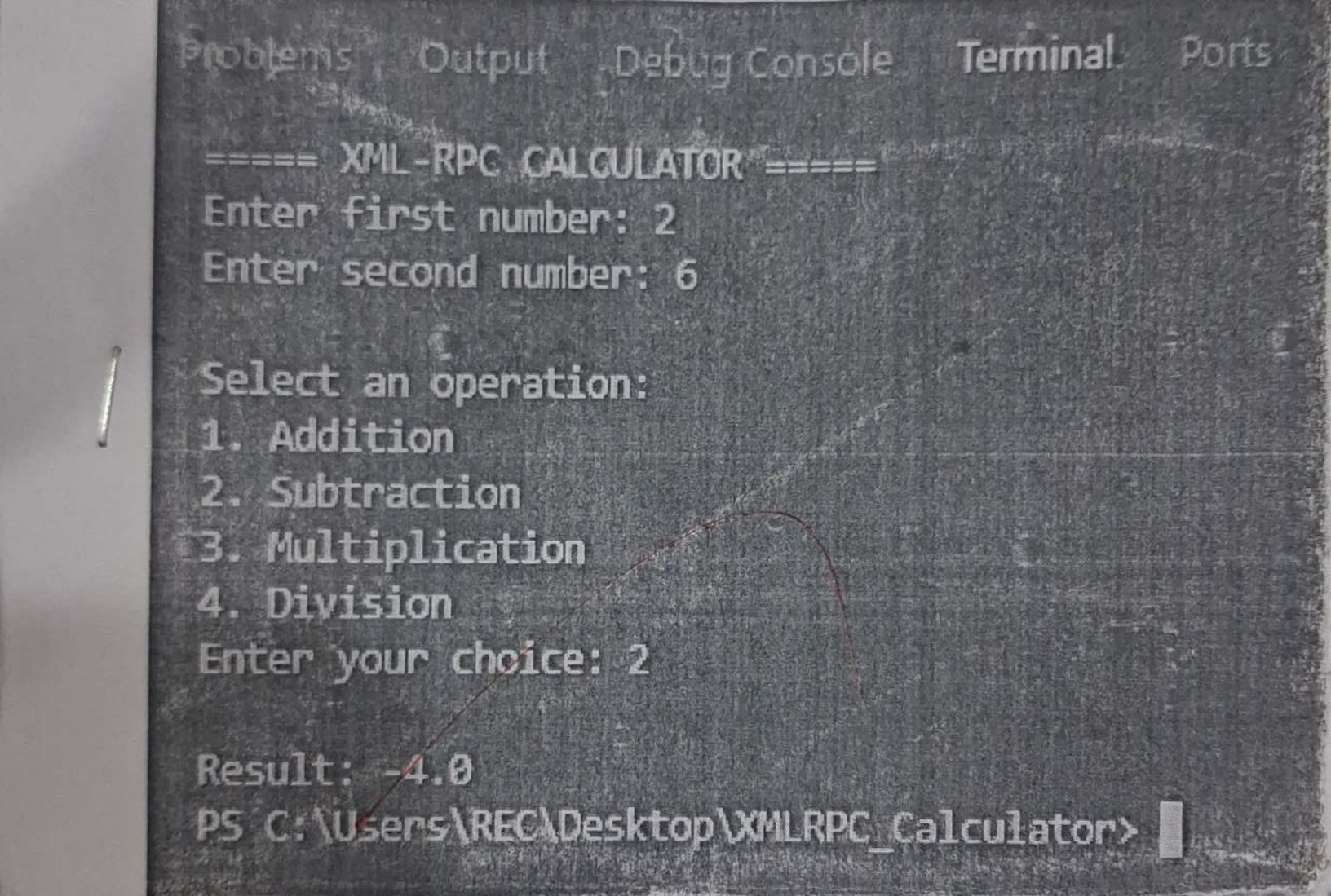
    def divide(self, x, y):
        if y != 0:
            return x / y
        else:
            return "Error: Division by zero is not allowed"

server = SimpleXMLRPCServer(("localhost", 8000))

print("XML-RPC server is running on port 8000...")

server.register_instance(Calculator())

server.serve_forever()
```


Problems Output Debug Console Terminal Ports
===== XML-RPC CALCULATOR =====
Enter first number: 2
Enter second number: 6
Select an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 2
Result: -4.0
PS C:\Users\REC\Desktop\XMLRPC_Calculator>

Client side code

```
import xmlrpc.client

proxy = xmlrpc.client.ServerProxy("http://localhost:8000/")

print("Addition: 10+5 =", proxy.add(10,5))

print("Subtraction: 10-5 =", proxy.subtract(10,5))

print("Multiplication: 10*5 =", proxy.multiply(10,5))

print("Division: 10/5 =", proxy.divide(10,5))

print("Division by zero: 10/0 =", proxy.divide(10,0))
```

output

XML-RPC server is running on port 8000...

client side output:

Addition: 10+5=15

Subtraction: 10-5=5

Multiplication: 10*5=50

Division: 10/5=2.0

Division by zero: 10/0 = Error: Division by zero is not allowed

Result:

The XML-RPC based calculator was successfully implemented. The client communicated with server. Using XML-RPC performed by arithmetic operation. Remotely, and display accurate results. This demonstrated use of XML-RPC protocol for remote function

Problems Output Debug Console Terminal Ports

```
===== XML-RPC CALCULATOR =====
Enter first number: 7
Enter second number: 4

Select an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 4

Result: 1.75
PS C:\Users\REC\Desktop\XMLRPC_Calculator>
```

Problems Output Debug Console Terminal

```
===== XML-RPC CALCULATOR =====
Enter first number: 5
Enter second number: 8

Select an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 3

Result: 40.0
PS C:\Users\REC\Desktop\XMLRPC_Calculator>
```

Exp No: 12

DATE: 24.9.2026

NMAP to discover live host using ARP scan, Icmp scan and tcp/Udp ping scan

Aim:

To discover live host in a network using Nmap with ARP, ICMP, TCP, and UDP ping scans using Try hack Me platform

Definition:

Nmap host discovery is used to identify active / live hosts in a network. It uses different probes such as ARP, ICMP, TCP and UDP to determine whether a host is reachable

ARP - Address resolution protocol

. ARP is used to find the MAC address of a device from its IP address within local network

ICMP -> Internet control message protocol

. ICMP is a network protocol used for error reporting and network protocol diagnostics. The ping command commonly uses ICMP Echo and Echo reply messages

TCP - Transmission control protocol

TCP is a connection-oriented transport protocol that provides reliable and ordered delivery of data between devices

UDP → User Datagram Protocol

UDP is a connectionless transport protocol that sends data like host establishing a connection It is faster but doesn't guaranteed delivery

Nmap:

Nmap is a network scanning tool used to discover live hosts, identify open ports, gather information about network services

Procedure:

[i] open the assigned tryHackme room and start the target machine

[ii] open the terminal provided in the tryHackme environment

[iii] Note the target IP address provided by tryHackMe

[iv] perform ARP, ICMP, TCP and NDP host discovery scan using Nmap

[v] observe the output and identify the hosts shown as "Host is up"

Nmap commands used

[i] General live Host discovery :-

command :nmap - Pn 10.49.151.126

observation :- The target host was detected as alive

[ii] ARP - Scan :-

command :- nmap :- sn - PR 10.49.151.126

observation : The ARP scan successfully detected the target as a live host

[iii] ICMP Scan :

command: nmap -sn - PE 10.49.151.126

observation : The ICMP scan detected the target as live host.

[iv] TCP - SYN Ping :

command: nmap -sn - PS 10.49.151.126

observation : The TCP SYN scan detected the target as a live host

[v] TCP ACK Ping

command: nmap sn - PA 10.49.151.126

observation :- The TCP ACK scan detected the target as a live host

[vii] UDP Ping:

Command: nmap -sn -PU 10.0.4.151.126

Observation:- The UDP scan detected the target as a live host

Result:

The discovery of live active host in a network using nmap by performing ARP scan I can use TCP/IP Ping scan in Try Hack Me Platform verified Successfully

Ex No: 6

DATE.

Building Anonymous FTP scanner using FTPLAB MODULE

AIM:

To create a python based FTP scanner using the ftplib module that:

* Checks if a public FTP server allows anonymous login.
* Lists the contents of directories on the server.

PROCEDURE:

* Import the built-in ftplib module.
* Write two python scripts:
* Sender side: Connects to the FTP server, logs in anonymously and lists the root directory.
* Receiver side: Connects anonymously and lists a specific sub-directory.
* Run the scripts in Google Colab or any Python environment.
* Observe the login status, server message, and directory contents.

CODE:

Vendor side code:

```
ftp lib
def anonymous login (host name):
    try:
        ftp = ftp lib.FTP (hostname)
            = ftp.login
        anonymous login

    Print (response)

    If "230" in response:
        + str (hostname) + "FTP Anonymous Login succ")
        Print ("welcome message: \n", ftp.getwelcome
        ())

        Print ("[-]" + str (hostname) + "FTP.
        Anonymous failed") Print ("Error", str(e))

        ftp 'ls.debian.org'
        anonymous login (hostname)
```

Receiver side code:

```
import ftp lib
    def check - anonymous - connection (host, path):
    try:
        with ftp.lib.FTP (host) as connection:
        connection.login ('anonymous', 'anonymous')
        Print (f"Listing directory: {path}:")
        connection.cwd (path) for name, details
        in connection.mlsd ():
```

- Print(name, details['type'], details.get('size', 'N/A'))

except Exception as e:

Print("Error", str(e))

FTP_server_URL = 'ftp.cse.cdlu.org'.

DOWNLOAD_PATH '/pub/linux/kernel/v5.x/'

check_anonymous_connection(FTP_server_URL, download_dir_path)

OUTPUT: Server side output:

230 Anonymous access granted, restrictions apply [+] ftp.cse.cdlu.org FTP-Anonymous Login Successful welcome message:

220 PRO FTPD Server (mirror.cse.cdlu.org) [::ffff:192.168.1.4]

Client side output:

drwxrwxr-x ftp ftp

Received data output:

Welcome: 220 PRO FTPD Server (mirror.cse.cdlu.org [::ffff:192.168.1.4])

Listing directory: /pub/linux/kernel/v5.x

linux-5.10.1.tar.xz file 113072504

linux-5.10.11.tar.xz file 113167464

RESULT:

* the script successfully connected to FTP server
* It verified anonymous login is allowed
* The directory contents were retrieved and displayed
* The Proves that the server provides public access and is working correctly

Ex No: 11

Date

Examine Network Address Translation using Cisco Packet Tracer.

Aim:

Implement Network Address translation using Cisco packet tracer will configure static dynamic and port Address translation on a router to enable communication between a private internal and a public external network

Procedure:

1. open Cisco packet tracer
2. Add devices:

   Drag PC0 and PC1 representing inside private network
3. connect devices:

Step 1: open Cisco Packet Tracer

Step 2: Add devices

- Add switches
- Add routers
- Add a cloud or Another Router

Step 3:

1. Connect PC1 and PC2 to the switch:

* Select copper straight-through cable & click on PC1, choose the FastEthernet port

2. Connect switch to Router

3. Connect Router to outside Network device

Step 4: Verify Connection

* The links should show green or orange when devices are connected properly

* PC1: 192.168.1.10/24

* PC2: 192.168.1.11/24

* Router GigabitEthernet 0/0: 192.168.1.1/24 outside Network:

* Router GigabitEthernet 0/1: 200.1.1.1/24 show of Router

Step 2: Assign IP Addresses

Step 1: Assign IP Address to PC1 and PC2

* Click on PC1 in the workspace

* Go to desktop tab

* Click on IP Configuration

* Enter the following

* IP 192.168.1.10 Subnet Mask 255.255.255.0

- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

2. Assign IP Address to Router interfaces

- click on the Router.
- click on the CLI tab

using the config tab

- Go to GigabitEthernet 0/0
- Set the IP Address: 192.168.1.1
- Subnet Mask: 255.255.255.0
- turn on the interface by checking Port status

do on

- Go to GigabitEthernet 0/1
- set the IP Address 200.1.1.1
- Subnet mask: 255.255.255.0
- Turn on the interface by checking Port status

do on using the CLI method:

enable configure terminal

3. Assign IP to outside device:

- If it is a Router
- IP to it a cloud
- click Home Go config tab
- Assign the IP address 200.1.1.2 with

Subnet mask 255.255.255.0

Step 4: Verify - IP configuration

A From PC1 and PC2 go to the desktop tab →
command prompt :.

A ping the router inside interface :

A From Router IP. any the outside device ping

200.1.1.2 step 3: open the router CLI and enter the following commands :

CODE:

Step 4: enable

configure

terminal

interface gigabit ethernet 0/0 ip

address 192.168.1.1 255.255.255.0 no shutdown

exit

interface gigabit ethernet 0/1 ip address

200.1.1.1 255.255.255.0

no shutdown

exit

4.1 configure inside and outside interfaces for

: configure terminal interface

gigabit ethernet 0/0 ip nat

inside exit interface.

gigabit ethernet 0/1 ip nat

outside exit

4.2 Static NAT configuration

4.3 Dynamic NAT Configuration

How to set Default Gateway on PC1 and PC2

1. Click on PC1 in the workspace
2. Go to the desktop tab
3. Click on IP configuration.
4. On the Default Gateway field type 192.168.1.1
5. Make sure the IP Address and Subnet mask are already set.
6. Close the window

6.1 Check the NAT translation

6.2 Test connectivity

RESULT:

The router successfully translated private IPs to public ones using static, dynamic, and PAT configuration

All inside host could communicate with the external network, confirming that NAT worked properly in packet tracer

Ex No. 13
Date:

Demonstrate Network Forensics using pcapxray tool

Aim:

To analyze and visualize network traffic from a PCAP file using the pcapxray tool for forensic investigation.

Steps to complete network forensics Experiment using pcapxray

Step 1: Prepare your computer

Make sure you have Python 3 installed

If you don't have Python:

Go to python.org and download it

Follow the installation instructions for your operating system

Step 2: Install pcapxray tool

1. Open Command Prompt or Terminal.
2. Type this command and press Enter
3. pip install pcapxray

This installs the pcapxray tool on your computer

Step 3: Get a PCAP file

Option A: Use a sample PCAP file.

Download a sample PCAP file from a website like

https://wiki.wireshark.org/SampleCaptures

Save the file on your computer

Option B: Capture your own traffic.

Open Wireshark.

Click on the network interface to start capturing.

Let it capture for 1-2 minutes.

Step 4: Run pcap xray on your PCAP file.

1. Put the PCAP file in a folder you can easily access.

2. Open command prompt terminal and navigate to that folder. Example:

3. cd path to your folder

4. Run the command:

5. pcap xray.py -f capture.pcap

Step 5: View the result:

After the command runs, it will create a dir named pcapxray.

Step 6: Analyze what you see

Look for the biggest nodes - these are the most active devices

RESULT:

Pcap Xray successfully generated a visual map of hosts & connections and protocols, helping identify suspicious or abnormal network activities

Ex No:4

Date:

TO CAPTURE, SAVE, AND ANALYZE NETWORK TRAFFIC ON TCP / UDP / IP / HTTP / ARP / DHCP / ICMP, DNS using wireshark tool

Aim:

The aim of the experiment is to capture, save and analyze network traffic on TCP / UDP / IP / HTTP / ARP / DHCP / ICMP / DNS using wireshark tool.

procedure:

Step 1: Download and install wireshark from its official website. ensure you have administrator privileges for network capture tool.

Step 2: Launch wireshark and identify the network interface through which your device is connected to the network.

Step 3: Before starting the capture, apply capture filters to focus on specific protocols and reduce unwanted data.

Example of capture filters you can use:

Capture only TCP traffic : tcp

Capture only UDP traffic : udp

Capture only HTTP traffic : tcp port 80

Capture only DNS traffic : dns

Capture only ARP traffic : arp

Capture DHCP traffic : port 67 or port 68

Step 4: Click the start button to begin capturing the network

Step 5: While capturing use display filters to isolate and analyze specific protocols more easily. Example,

To display only UDP packets: udp

Step 6: Let the capture run until you have collected enough packets

Step 7: Stop the captured traffic by navigating

Step 8: Review and analyze the captured packets in detail. Look for key information

output:

The output will be list of network packets displayed in wireshark's packet details window which will include.

Sample output:

No.	Time	Source	Destination	Protocol	
1	0.00000	192.168.1.1	8.8.8.8	ICMP	
2	0.002748	8.8.8.8	192.168.1.1	ICMP	64 Echo Reply
3	0.003736	192.168.1.1	192.168.1.1		

RESULT:

Thus the experiment capture done and analyze network traffic on TCP/UDP/IP/HTTP/DNS

Expt No:
Date:

Analyze the different types of servers using
- Webalizer TOOL

AIM:

To analyze the usage statistics of different type of servers using webalizer tool

PROCEDURE:

1. Install the webalizer tool in the system
Sudo apt-get update sudo
apt-get install webalizer
2. verify installation
3. Locate the log files of the server.
4. Run webalizer on the log file to Generate a report.
5. Open a web browser and open the report
6. Repeat the same process
7. Compare and analyze the reports

PROGRAM:

Step 1: Install webalizer sudo apt-get install webalizer

Step 2: Run webalizer

/var/log/apache2/access.log.0/var/www/html/apache_report/

Step 3: Run webalizer for Nginx logs sudo webalizer

/var/log/nginx/access.log.0.

Step 4: Run Webalizer for proxy logs sudo webalizer

./var/log/squid/access.log.0

Sample observation Table

- Server
- Type
- Total
- Hits
- Unique
- Visitors
- Bandwidth
- Used
- Top

Client / Domain Report URL

Apache 12,560 1240 1.8 GB 192.168.1.10.

Pie chart of top hosts and URLs

RESULT:

The webalizer tool successfully generated. Graphical and statistical reports of different servers Total hits and visits

Bandwidth usage

TOP IP's and URL's

Daily monthly and hourly traffic

Exp No: 8

Date

Develop a program to create reverse shell using TCP socket

Aim:

To develop a simple chatting application using Python socket programming to exchange messages between a client and a server.

Observation

Chatting allow users to exchange message through a computer network.

Procedure:

1. Install Python and open VS code
2. Create two files server.py and client.py
3. Import the socket and threading modules
4. Create a TCP socket in both program
5. Bind and start the server, then connect the client
6. Run the server first and the client next
7. Exchange messages between the two terminals
8. Type exit to end the chat.

Program:

Server program:

```
import socket
import threading

server = socket.socket()
server.bind(("127.0.0.1", 12345))
server.listen()
print("waiting for client...")
conn, addr = server.accept()

def receive():
    while true:
        msg = conn.recv(1024).decode()
        if not msg or msg.lower()=="exit"
        print("client", msg)

threading.Thread(target = receive, daemon
                                =true).start()

while true:
    msg = input("server:")
    conn.sendall(msg.encode())
    if msg.lower == 'exit':
        break

conn.close()
server.close()
```

Client Side:

```
client = socket.socket()
client.connect(("127.0.0.1", 12345))
print("Connected to server")
def received
    while True:
        msg = client.recv(1024).decode()
        if not msg or msg.lower() == "exit":
        print("server:", msg)

while True:
    msg = input("client: ")
    client.sendall(msg.encode())
    if msg.lower() == "exit":
        break
client.close()
```

output:

Server:	Client:
waiting for client	connected to server:
client connected	client: Hi
server: Hello	Server: Hello
client: Hi	client: I am fine
server: How are you?	server: How are you
client: I am fine	

Result :

This a Simple chatting application was implemented successfully using Python socket programming and TCP to exchange message between the client and server.